Boundaries and 'Not Sure'

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected uncertainty-and-review record.

Describe unclear inputs, limit stable predictions, request missing units and challenge unsupported review.

Keep one habit: Keep the original record, requested evidence and eventual outcome together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Save New Examples for Testing

Learning from Examples

If/Then Decisions

W1 · Identify the input uncertainty

> Compare the two predictions and request evidence about the smudged measurement.

Your evidence record

- For a smudged height that could be 5 or 6 cm, state what is known, what is unknown and the two **prediction**s under tall at least 6. Choose a check that could distinguish the heights.

  Explain why rerunning the threshold comparison cannot do that.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Limit a stable prediction

> Keep the exact-value uncertainty visible without inventing probabilities.

Your evidence record

- Assume the only possible heights are 8 and 9 cm. Apply tall at least 6 to each. Write a justified result statement and two things it does not establish. Do not invent a confidence percentage.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Request the missing unit

> Use source evidence rather than assigning the model's required unit.

Your evidence record

- A record says 60 but omits the unit. A reviewer suggests reading it as 60 cm because the model needs cm. Explain the problem, name the **evidence** to request and state the outcome if the unit cannot be established.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Inspect an unsupported review conclusion

> Distinguish model agreement from evidence about the original input.

Your evidence record

- A reviewer cannot read the original tower card but chooses 6 cm because two models both predict tall when given 6. Explain why this fails to resolve the **input**. Write a more defensible review record and a next step.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Transfer the method to distance

> Keep one first attempt before hints or feedback.

Read and explain independently

A paper model predicts far at a distance of at least 4 units, otherwise near, for whole-number distances 1 through 8. For each complete possibility set, compare **prediction**s: 2 or 3; 3 or 4; 4 or 5.

Identify the set requiring review before a single category can be stated. Design a relevant check and an **unresolved** ending. Limit the claims for the other two sets.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Transfer the method to distance

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Add a not-sure branch, missing-information prompt, and human review destination to the sorter.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Keep a stable **prediction** within its limits for **Uncertainty** and review. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Uncertainty** and review. Proposed explanation: “Two possible values imply equal probabilities”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A sorter receives a clear purple card, a red card hidden by a sleeve and a card labelled both red and blue by two reviewers. Give a different reason for **uncertainty** in each case and a suitable next action.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A sorter handles clearly red and clearly blue cards. Purple and very dark cards arrive.

How should it handle each uncertain case?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A sorter knows pencils and books. You show a lunch box. Should it force a known label or ask for help? State a useful fallback.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 16 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, **boundary case** or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A sorter receives a clear purple card, a red card hidden by a sleeve and a card labelled both red and blue by two reviewers. Give a different reason for **uncertainty** in each case and a suitable next action.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A paper model predicts far at a distance of at least 4 units, otherwise near, for whole-number distances 1 through 8. For each complete possibility set, compare **prediction**s: 2 or 3; 3 or 4; 4 or 5.

Identify the set requiring review before a single category can be stated. Design a relevant check and an **unresolved** ending. Limit the claims for the other two sets.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

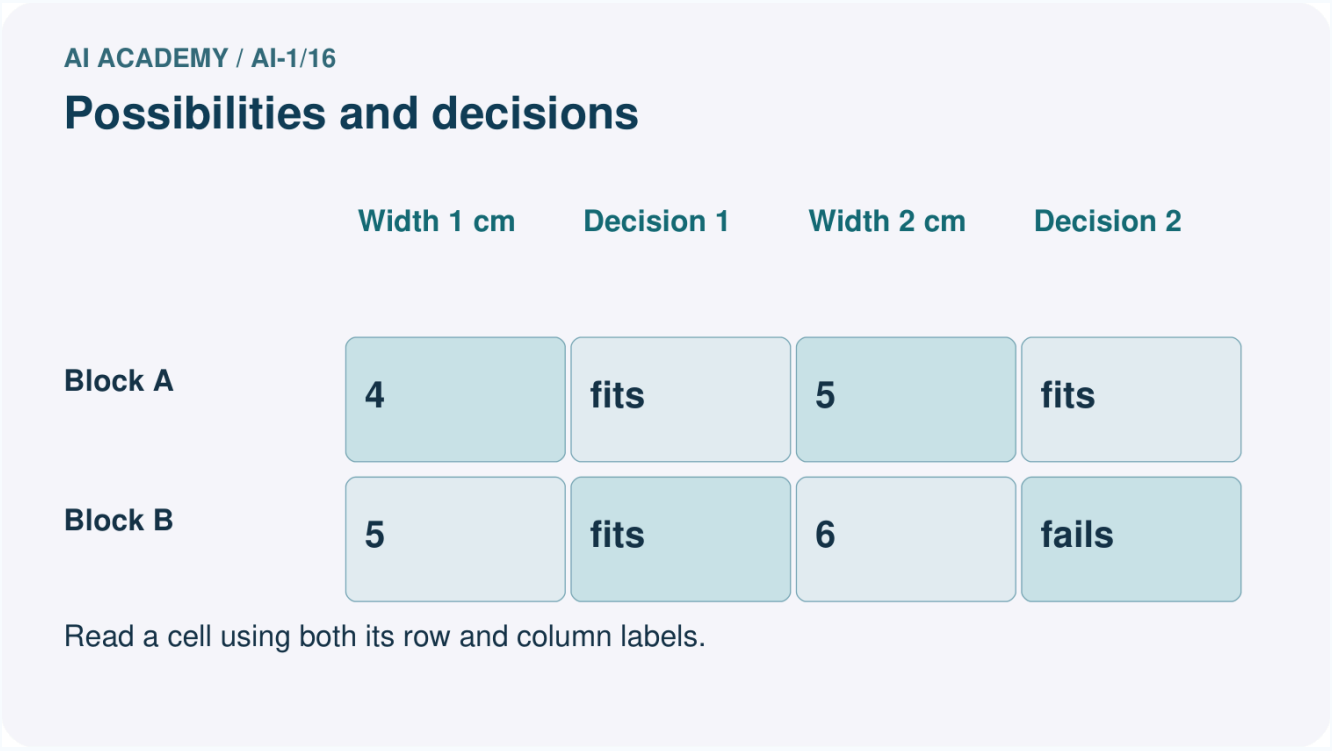
> Keep the explanation and the support you used.

Use this when helpful

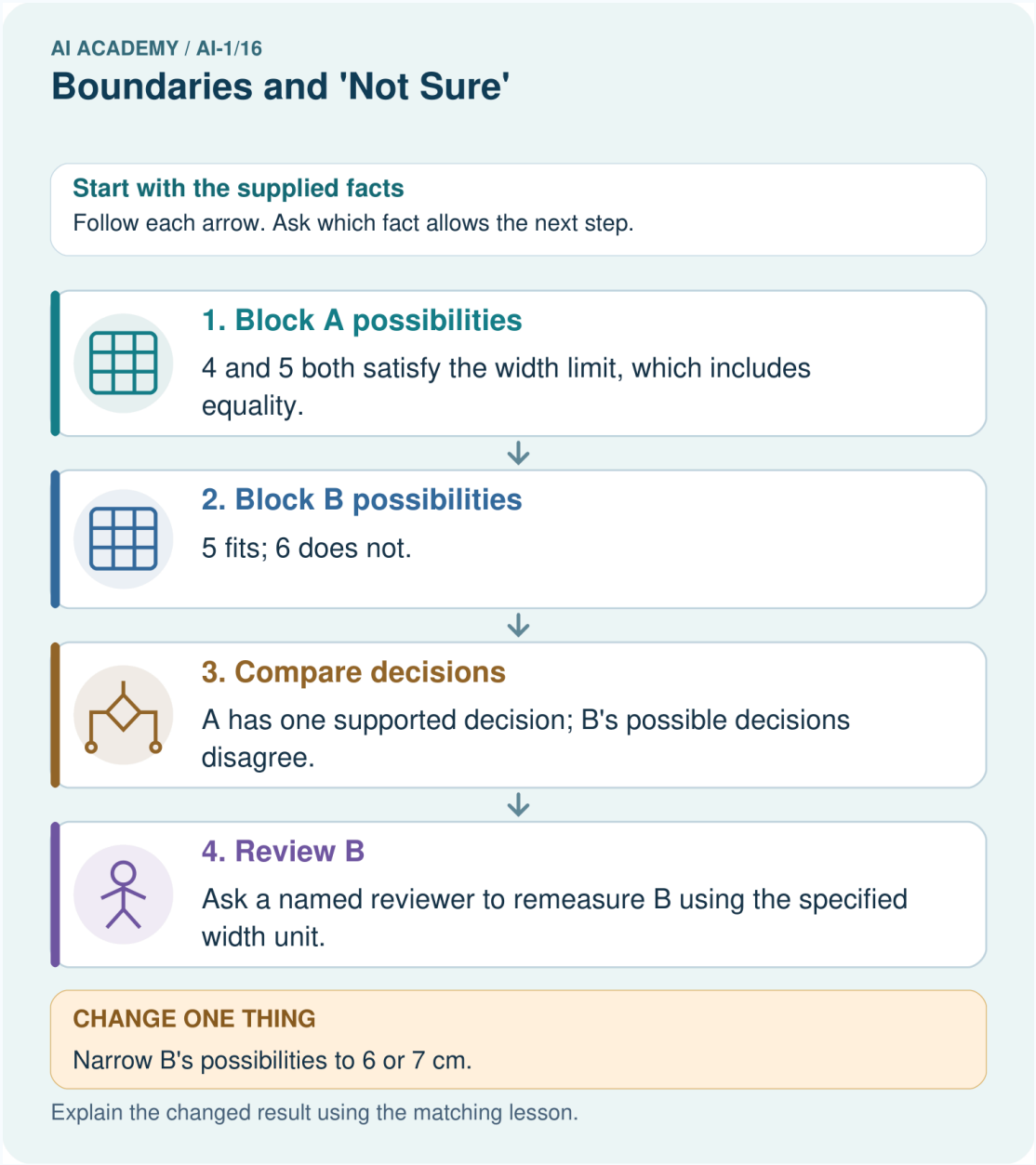
A fictional cubby selector accepts a block if width is at most 5 cm.

An unclear measurement means Block A is either 4 or 5 cm; Block B is either 5 or 6 cm.

These complete possibility sets contain all widths under consideration. Other dimensions are known to fit.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Which original block needs a better width before accepting or rejecting it? What should the reviewer do?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Add explicit review paths to the sorter, including **unresolved input**, responsible person and **evidence** needed.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Add explicit review paths to the sorter, including **unresolved input**, responsible person and **evidence** needed.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional cubby selector accepts a block if width is at most 5 cm.
- Which labelled step matches this detail? Ask a named reviewer to remeasure B using the specified width unit.
- 1. Block B possibilities
- 2. Compare decisions
- 3. Block A possibilities
- 4. Review B
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Review B Ask a named reviewer to remeasure B using the specified width unit. **Uncertainty** about a number need not always create **uncertainty** about the decision.

Every allowed width for A fits, so its width **uncertainty** does not affect acceptance. B straddles the boundary.

Record both possibilities and request a better measurement instead of selecting the one that makes a preferred outcome.

Changed-case question

- Changed-case practice: Change B's possibilities to 6 or 7 cm. Which conclusion is supported by the supplied facts?
- 1. Both fail the width limit; rejection no longer depends on resolving their difference.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Both fail the width limit; rejection no longer depends on resolving their difference. All remaining possibilities lead to the same decision.

Repair and save your work

Review Save New Examples for Testing (ai-1/15). Goal: Separate examples used to choose a model from examples reserved for its final evaluation. Short practice: Does 4/4 on the replay replace the original 3/4 result? Explain..

Return to this week and explain what you changed.

Review Learning from Examples (ai-1/13). Goal: Fit a threshold by tracing **prediction**s and counting disagreements for every candidate. Short practice: Which new measured length would reveal disagreement between the candidates?.

Return to this week and explain what you changed.

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which **input** detects the difference between the two written conditions?

Explain its entire run.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

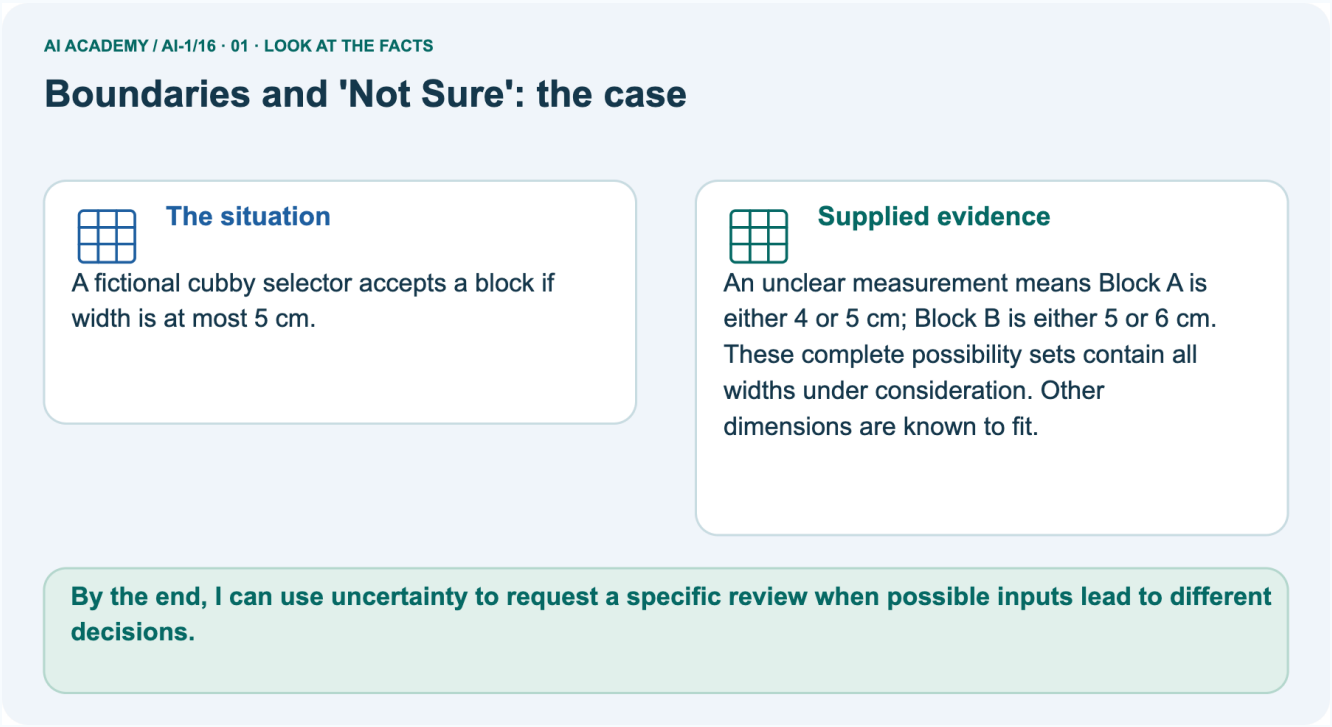
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Boundaries and 'Not Sure': the case



Follow the labelled picture with the matching explanation. Open the vector diagram

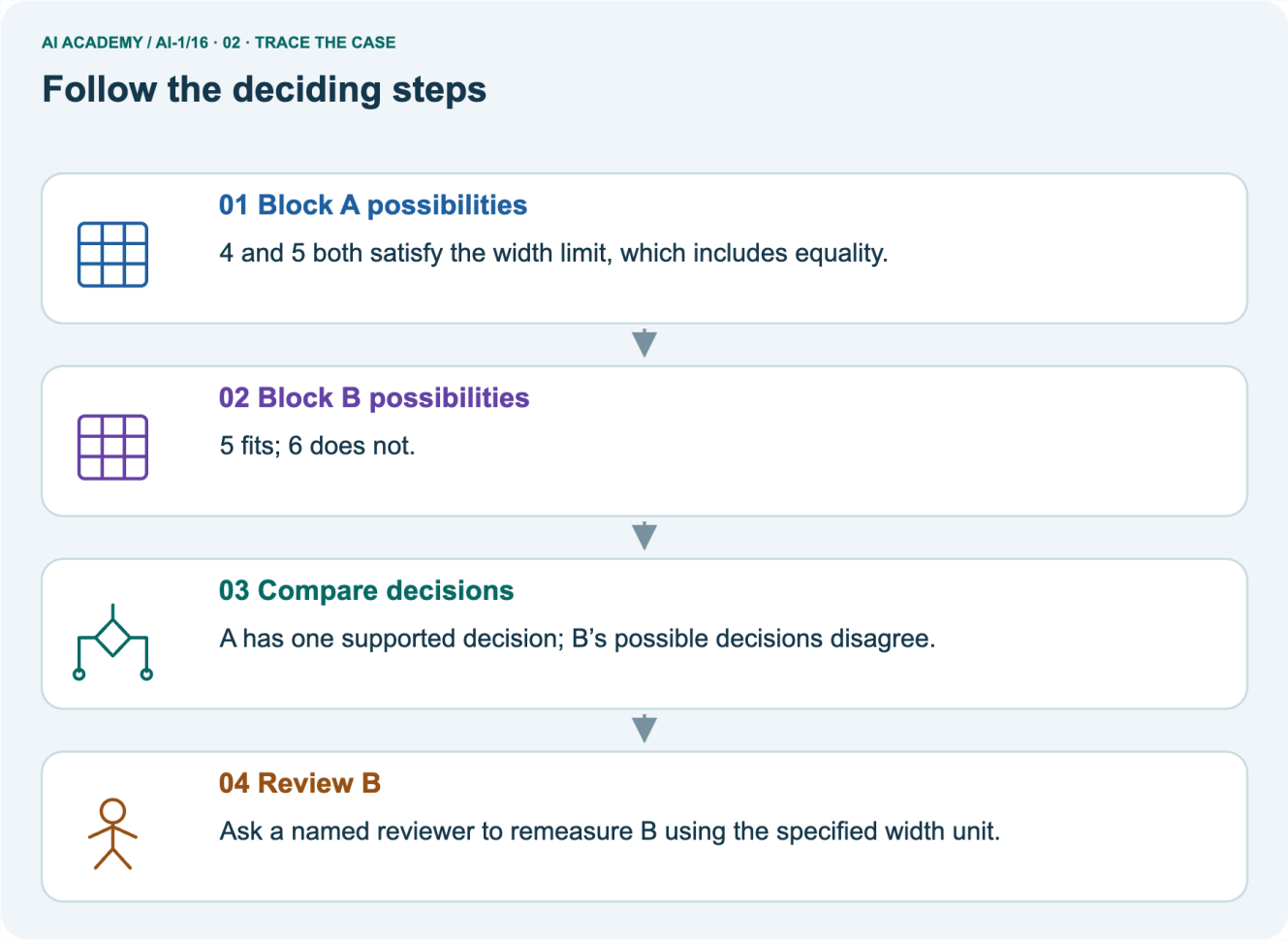
A fictional cubby selector accepts a block if width is at most 5 cm. Supplied facts: An unclear measurement means Block A is either 4 or 5 cm; Block B is either 5 or 6 cm.

These complete possibility sets contain all widths under consideration. Other dimensions are known to fit.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



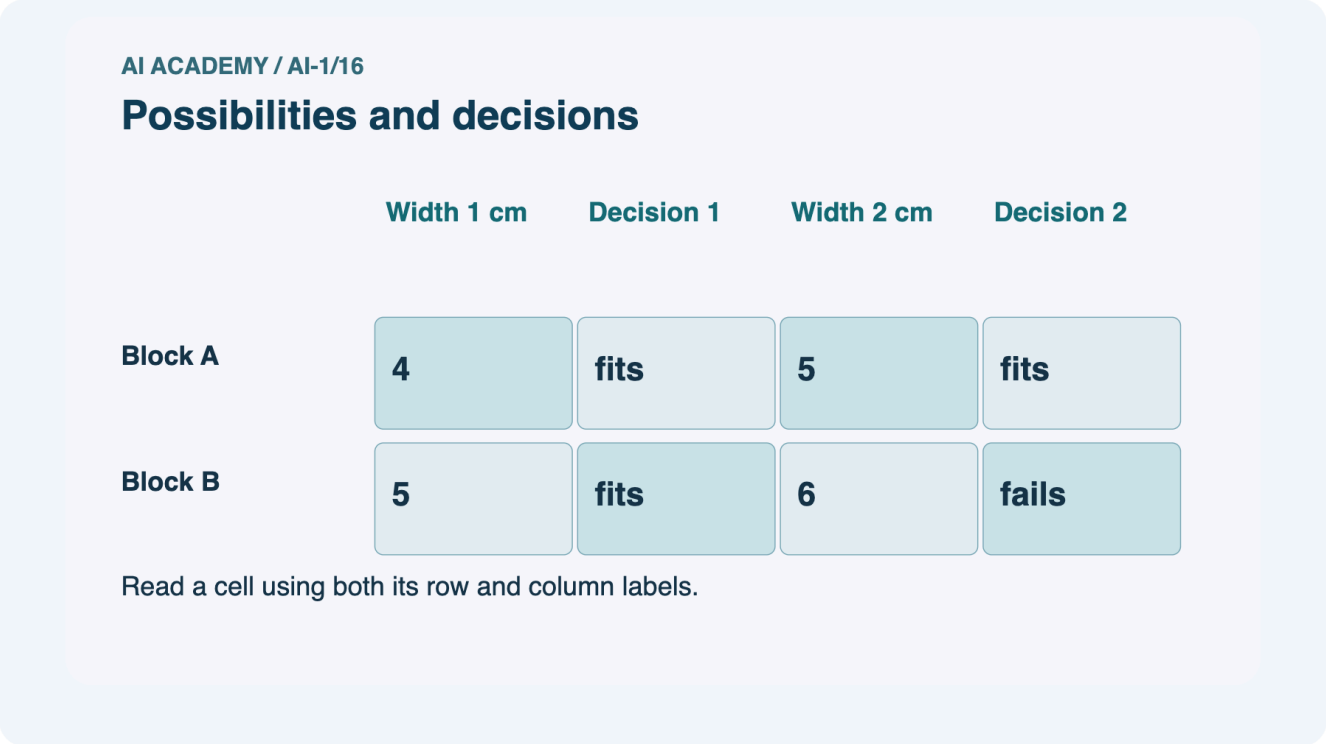
Follow the labelled picture with the matching explanation. Open the vector diagram

Block A possibilities: 4 and 5 both satisfy the width limit, which includes equality. → Block B possibilities: 5 fits; 6 does not. → Compare decisions: A has one supported decision; B's possible decisions disagree. → Review B: Ask a named reviewer to remeasure B using the specified width unit.

Point to each numbered stage and explain which recorded **input** it uses.

> **My picture explanation, deciding evidence and one limit**

03. Possibilities and decisions



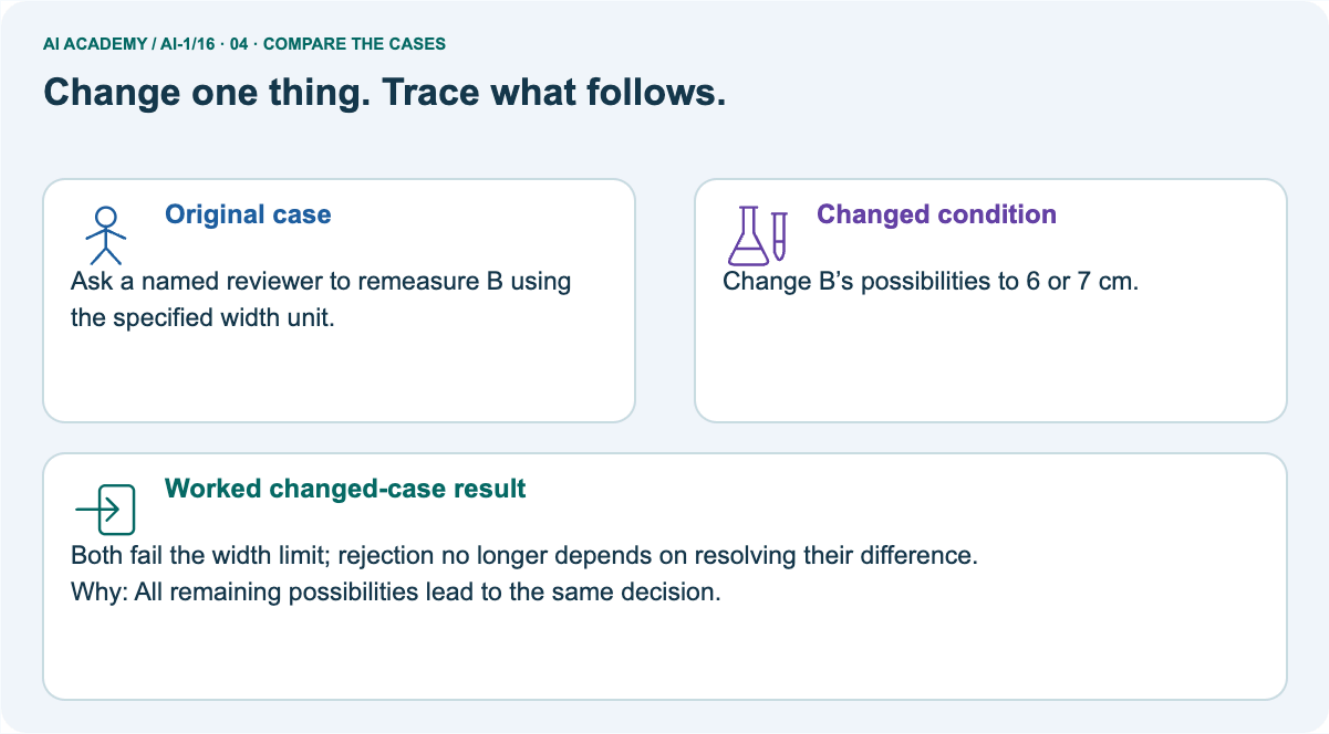
Follow the labelled picture with the matching explanation. Open the vector diagram

Possibilities and decisions. Block A: Width 1 cm = 4, Decision 1 = fits, Width 2 cm = 5, Decision 2 = fits. Block B: Width 1 cm = 5, Decision 1 = fits, Width 2 cm = 6, Decision 2 = fails.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

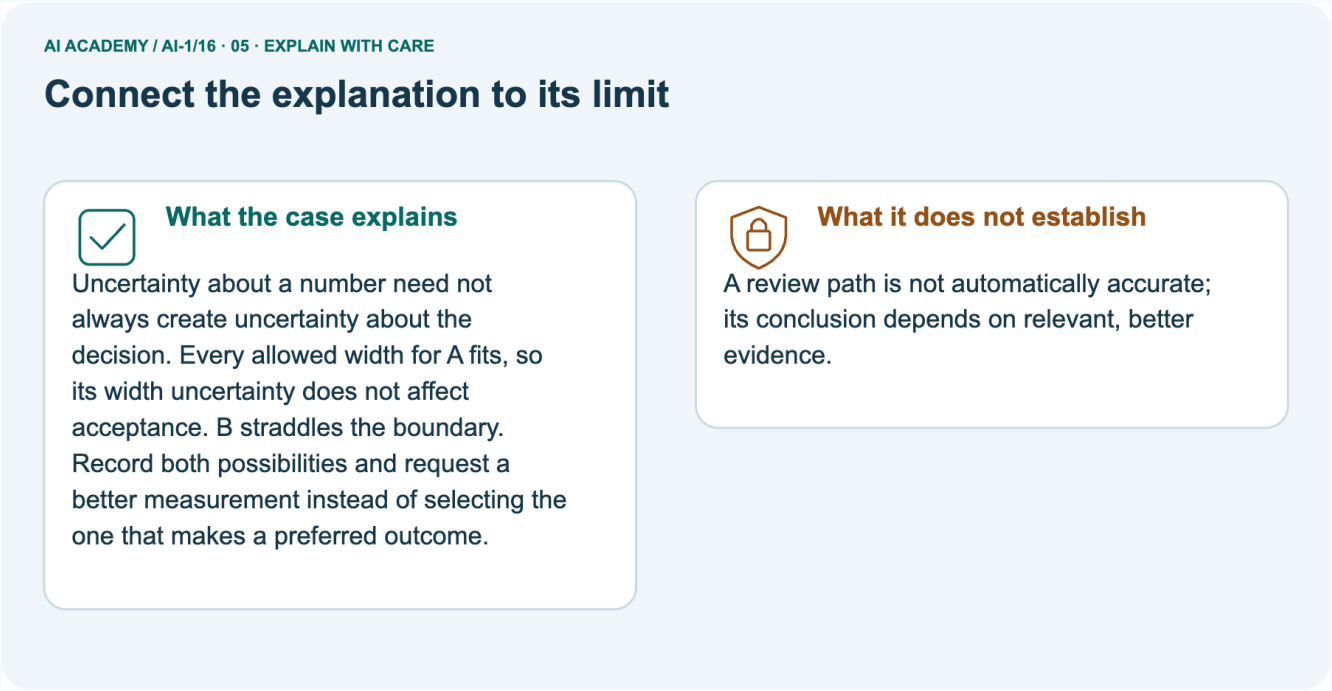
Original case: Ask a named reviewer to remeasure B using the specified width unit. Changed condition: Change B's possibilities to 6 or 7 cm.

Expected result: Both fail the width limit; rejection no longer depends on resolving their difference. Reason: All remaining possibilities lead to the same decision.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

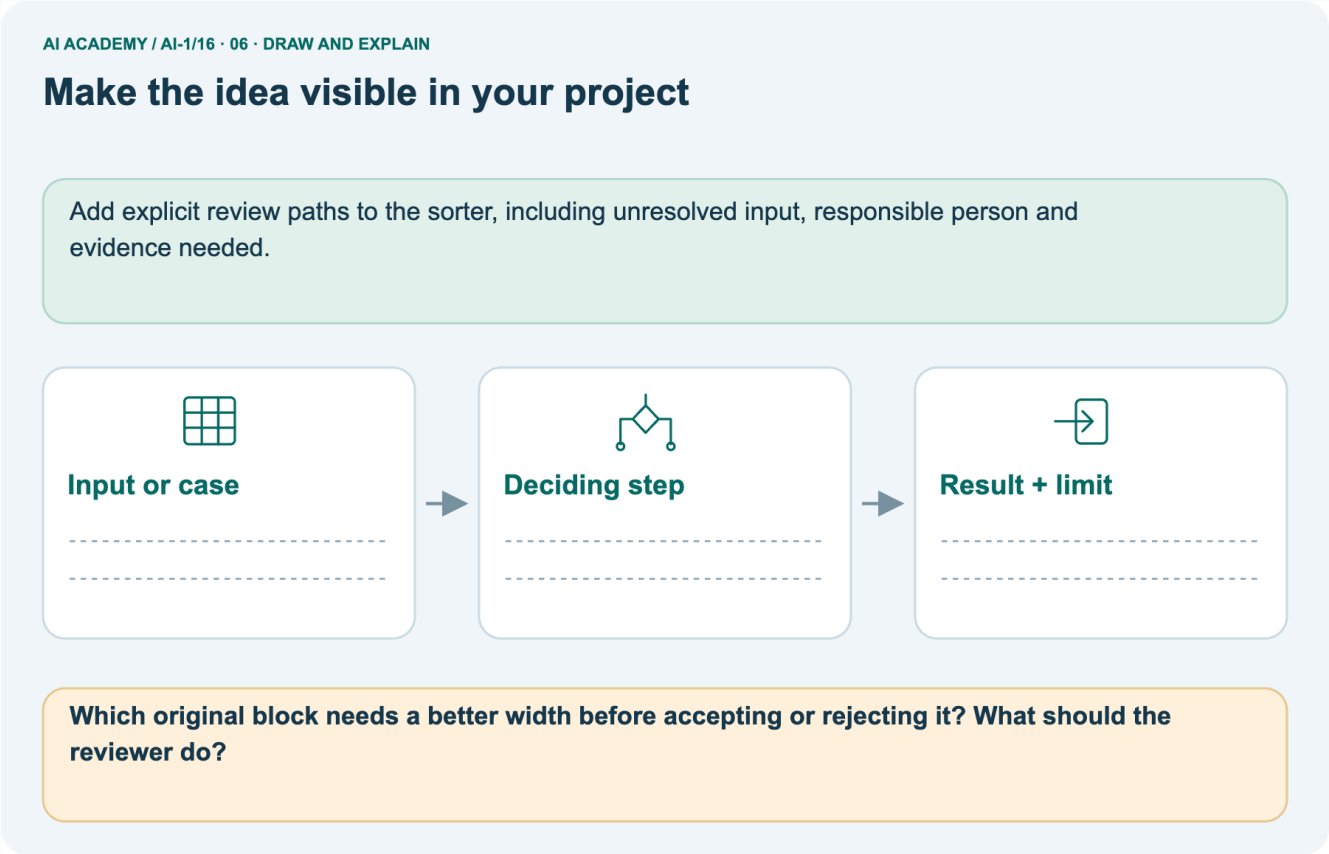
Uncertainty about a number need not always create **uncertainty** about the decision. Every allowed width for A fits, so its width **uncertainty** does not affect acceptance. B straddles the boundary.

Record both possibilities and request a better measurement instead of selecting the one that makes a preferred outcome. Limit: A review path is not automatically accurate; its conclusion depends on relevant, better **evidence**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Add explicit review paths to the sorter, including **unresolved input**, responsible person and **evidence** needed. Draw your **input**, deciding step, result and limit.

Supported question: Which original block needs a better width before accepting or rejecting it? What should the reviewer do?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Boundaries and 'Not Sure': the case

02. Follow the deciding steps

03. Possibilities and decisions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.